

DNS Giữa 2 máy Window

I. Cấu hình 2 máy

1. Máy Server : Host-Only tắt DHCP tránh nhận IP từ VirtualBox, IP tĩnh : 192.168.56.10
2. Máy Client : Host-Only tắt DHCP tránh nhận IP từ VirtualBox

II. Cài đặt DNS

1. Trên máy Server:

```
Install-WindowsFeature DNS -IncludeManagementTools
Get-WindowsFeature DNS*
```

```
$ZoneName = "lab.demo"
```

```
$ReverseZoneName = "56.168.192.in-addr.arpa"
```

```
Add-DnsServerPrimaryZone `
```

```
-Name $ZoneName `
```

```
-ZoneFile "$ZoneName.dns"
```

-> Tạo Zone Name

```
Add-DnsServerPrimaryZone `
```

```
-Name $ReverseZoneName `
```

```
-ZoneFile "$ReverseZoneName.dns"
```

-> tạo Reverse Zone Name

```
Get-DnsServerZone
```

-> lấy ra danh sách Zone Name

```
Add-DnsServerResourceRecordA `
```

```
-ZoneName $ZoneName `
```

```
-Name "server" `
```

```
-IPv4Address "192.168.26.10"
```

-> Add 1 record loại A (theo địa chỉ IP) với tên: server vào IP : 192.168.56.10. Hay : trở server.lab.demo vào ip : 192.168.56.10

```
Get-DnsServerResourceRecord `
```

```
-ZoneName "lab.demo" `
```

```
-RRType NS
```

-> Check lại danh sách Record trong Zone name trên

```
Add-DnsServerResourceRecordPtr `
```

```
-ZoneName $ReverseZoneName `
```

```
-Name "10" `
```

```
-PtrDomainName "server.lab.demo"
```

-> Add PTR Record cho Server, hiểu đơn giản là trở ip 192.168.56.10 vào server.lab.demo

III. Kiểm tra

Do trên máy Client mình có cài DHCP rồi, mình chỉ cần sửa DHCP Option để **DHCP cấp**

DNS Server = 192.168.56.10 cho client trong scope 192.168.56.0

```
Set-DhcpServerv4OptionValue `
```

-ScopeId "192.168.56.0" `

-DnsServer 192.168.56.10

Get-DhcpServerv4OptionValue -ScopeId "192.168.56.0" -> check lại

1. Trên máy client :

gõ

ipconfig /release

ipconfig /renew

ping server.lab.demo -> Nó sẽ tự hiểu ping đến 192.168.56.10 và có reply success

```
Link-local IPv6 Address . . . . . : fe80::9c6:93b6:727e:5fbd%5 (Preferred)
IPv4 Address. . . . . : 192.168.56.100(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Monday, May 25, 2026 9:06:53 AM
Lease Expires . . . . . : Tuesday, June 2, 2026 9:06:52 AM
Default Gateway . . . . . :
DHCP Server . . . . . : 192.168.56.10
DHCPv6 IAID . . . . . : 101107623
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-A5-39-88-00-00-27-C4-75-D6
DNS Servers . . . . . : 192.168.56.10
NetBIOS over Tcpip. . . . . : Enabled
PS C:\Users\Administrator> ping server.lab.demo

Pinging server.lab.demo [192.168.56.10] with 32 bytes of data:
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time=1ms TTL=128
Reply from 192.168.56.10: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.56.10:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
Control-C
PS C:\Users\Administrator>
```

2. Trên máy Server:

```
PS C:\Users\Administrator> nslookup 192.168.56.10 127.0.0.1
Server: localhost
Address: 127.0.0.1

Name:      server.lab.demo
Address: 192.168.56.10

PS C:\Users\Administrator> nslookup server.lab.demo 127.0.0.1
Server: localhost
Address: 127.0.0.1

DNS request timed out.
    timeout was 2 seconds.
Name:      server.lab.demo
Address: 192.168.56.10

PS C:\Users\Administrator> Get-DnsServerResourceRecord -ZoneName $ZoneName -RRType A

HostName                RecordType Type      Timestamp
-----
server                   A          1          0
```

IV. Edit DNS Record

```

$Old = Get-DnsServerResourceRecord `
    -ZoneName $ZoneName `
    -Name "server" `
    -RRType A

$New = $Old.Clone()
$New.RecordData.IPv4Address = "192.168.56.100"

Set-DnsServerResourceRecord `
    -ZoneName $ZoneName `
    -OldInputObject $Old `
    -NewInputObject $New

```

Mục đích sửa Record lại A từ ip : 192.168.56.10 sang ip : 192.168.56.100

Test lại :

```

PS C:\Users\Administrator> nslookup server.lab.demo 127.0.0.1
Server: localhost
Address: 127.0.0.1

Name:    server.lab.demo
Address: 192.168.56.100

```

V. Add MX Record

```

Add-DnsServerResourceRecordA `
    -ZoneName $ZoneName `
    -Name "mail" `
    -IPv4Address "192.168.56.20"

```

-> Trỏ mail.lab.demo gắn với ip : 192.168.56.20

```

Add-DnsServerResourceRecordMX `
    -ZoneName $ZoneName `
    -Name "@" `
    -MailExchange "mail.lab.demo" `
    -Preference 10

```

-> Add Mx Record : mail server đặt trên domain gốc

Kết quả:

```

PS C:\Users\Administrator> Add-DnsServerResourceRecordA `
>> -ZoneName $ZoneName `
>> -Name "mail" `
>> -IPv4Address "192.168.56.20"
PS C:\Users\Administrator> Add-DnsServerResourceRecordMX `
>> -ZoneName $ZoneName `
>> -Name "@" `
>> -MailExchange "mail.lab.demo" `
>> -Preference 10
PS C:\Users\Administrator> Get-DnsServerResourceRecord -ZoneName $ZoneName -RRType MX

```

HostName	RecordType	Type	Timestamp	TimeToLive	RecordData
@	MX	15	0	01:00:00	[10][mail.lab.demo.]

```

PS C:\Users\Administrator>

```

```

PS C:\Users\Administrator> Get-DnsServerResourceRecord -ZoneName $ZoneName -RRType A

```

HostName	RecordType	Type	Timestamp	TimeToLive	RecordData
mail	A	1	0	01:00:00	192.168.56.20
server	A	1	0	01:00:00	192.168.56.100

VI. Add TXT Record

```

PS C:\Users\Administrator> Add-DnsServerResourceRecord `
>> -ZoneName $ZoneName `
>> -TXT `
>> -Name "@" `
>> -DescriptiveText "Toi ten la PhamVanQuyen"
PS C:\Users\Administrator> Get-DnsServerResourceRecord -ZoneName $ZoneName -RRType TXT

```

HostName	RecordType	Type	Timestamp	TimeToLive	RecordData
@	TXT	16	0	01:00:00	Toi ten la PhamVanQuyen